

Sam Crouch

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EDUCATION

Missouri University of Science and Technology

Rolla, MO

B.S. Computer Science

May 2027

Relevant Coursework: Data Structures, Algorithms, Operating Systems, Programming Languages, Microcontrollers and Embedded Systems, File Structures and Database Systems, Introduction to Artificial Intelligence, Cryptography, Introduction to Computer Networks, Computer Security

EXPERIENCE

Controls Intern

May 2026 – Present

Altec

Saint Joseph, MO

- Developed a Qt/QML library for embedded Linux displays, standardizing UI components across multiple product lines
- Implemented a C++ daemon to expose hardware functionality through a text file interface, decoupling application logic from low-level hardware control

Forklift Operator

May 2025 – Aug. 2025

Menards

Saint Peters, MO

- Operated forklifts in a lumber yard and warehouse
- Maintained product inventory and order fulfillment

Lube Tech

Nov. 2022 – Aug. 2024

Valvoline Instant Oil Change

Saint Peters, MO

- Performed preventative maintenance services including oil changes, fluid replacements, and filter installations
- Conducted vehicle inspections and communicated maintenance recommendations to customers

PROJECTS

pyVPW | *Python*

Sep. 2025 – Present

- Developed an open-source Python library implementing the SAE J1850 VPW protocol for GM vehicle diagnostics
- Designed the API to simplify development of diagnostic and data-logging tools
- Implemented the GM seed-key security algorithm to unlock PCMs, enabling writes to protected memory for tuning and diagnostic applications
- Published the project on GitHub with documentation and example applications, supporting adoption by other developers

LLM-TUI | *Python, Textual*

Feb. 2026 – May 2026

- Built a customizable, terminal-based AI assistant using Python and the Textual framework
- Designed an asynchronous terminal UI supporting real-time streaming responses, on-the-fly model switching, file attachments, and a snipping-tool style screenshot tool
- Integrated multiple LLM providers including OpenAI, Anthropic, Deepseek, Grok, and Ollama behind a common interface, enabling seamless provider switching

TECHNICAL SKILLS

Languages: C, C++, Python, QML, Bash, SQL, HTML/CSS

Frameworks: Qt, Textual, Django

Developer Tools: Git, Docker, CMake, Conan, Linux, GDB

Embedded/Automotive: Embedded Linux, CAN Bus, UART, AVR Microcontrollers